

Pietro Saggionetto

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EDUCATION

Politecnico di Milano

2023 – Present

Master's degree in mechanical engineering - Computational Mechanical Design | 27.7/30

Milan, IT

- **Key topics:** Turbomachinery, Computational Fluid Dynamics, Finite Element Modeling, Dynamics of Mechanical Systems, Topology Optimization, Digital Twin, Machine Learning
- **Thesis:** Design and CFD Validation of a Two-Stage Low-Pressure Marine Turbine within the European MARPOWER Project; end-to-end development from meanline design to full 3D geometry generation, with aerodynamic validation performed using ANSYS CFX

Università degli Studi di Padova

2020 – 2023

Bachelor's degree in mechanical engineering | 103/110

Padova, IT

- **Thesis:** FEM analysis of stress concentrations and deformation - comparison between traditional and meshless solvers

ITI Severi

2014 – 2020

Mechanics, Mechatronics, and Energy | 98/100

Padova, IT

EXPERIENCE

Race UP Team - Università di Padova (Formula Student)

2021 – 2023

Powertrain Engineer - Combustion Vehicle (ICE)

Italy

- Designed and manufactured in-house a 7-liter **carbon-fiber fuel tank** (layup-to-curing process)
- Developed the exhaust system including implementation of an **EXUP valve** configuration
- Managed engine control unit (LifeRacing ECU) activities and contributed to **engine mapping calibration** on dyno
- Took part in vehicle assembly, dynamic testing, and all competitions in years 2022 and 2023

Workshop Placements - Mechanical Companies & Erasmus+

2018 – 2021

Industrial workshop experience

Italy / United Kingdom

- Approximately **700 hours of hands-on workshop experience** in 4 different mechanical companies
- One-month **Erasmus+ industrial placement** in the United Kingdom (2019) at Winward Engineering (Walsall, UK)
- Conventional machining: **turning, milling, and surface grinding**, with **basic CNC exposure** (workpiece handling, machine zeroing, and G-code fundamentals)
- Mechanical assembly and dimensional inspection using calipers, micrometers, dial indicators, and gauge blocks

SKILLS

CAD : SolidWorks, PTC Creo, Rhino Grasshopper

FEA : Abaqus, ANSYS Mechanical

CFD : ANSYS CFX, OpenFOAM

Programming : Python, MATLAB, Latex

Languages : Italian (native), English (B2 – TOEFL iBT 82/120, 2023)

General : Microsoft Office Suite, Driving License (B)

EXTRACURRICULAR ACTIVITIES

Activities : Youth animator in Catholic Action (Azione Cattolica), responsible for organizing and supporting activities for children and adolescents

Interests : Rock climbing, table tennis, gym training